

CURRICULUM VITAE: IAN FREDERICK JONES
SEG biographical WIKI page: https://wiki.seg.org/wiki/Ian_Jones
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PROFESSIONALLY RELATED EMPLOYMENT:

- 2021 – present: Consultant Senior Geophysical Advisor, Brightskies Geoscience
2013 – 2021: Distinguished Advisor, ION
2000 – 2012: Senior geophysical advisor, EAME, ION GX Technology
1999 – 2000: Manager, CGG corporate depth imaging R&D, and technology.
1995 – 1999: Manager, corporate depth imaging technology, responsible for imaging strategy and methodology within CGG, including co-ordination and support of imaging processing groups in all CGG centres.
1993 – 1995: Senior research geophysicist, CGG London: involved in developing velocity model building techniques for 3D depth migration. Supervisor of the London technical projects group, which performs specialized processing, including 3D pre-stack and post-stack depth migration, 3D AVO, and 3D tomographic inversion. Responsible for technical marketing and client support in special processing.
1991 – 1993: Senior research scientist, CGG France: involved in developing steep dip 3D depth and time migration algorithms.
1985 – 1991: Research and development geophysicist with the Compagnie Generale de Geophysique, London. Primarily responsible for seismic software development (specifically: convolutional and wave equation multiple suppression techniques; attenuation compensation; and enhanced stacking), programming for the CRAY-1S, XMP and YMP, as well as CDC machines. Also responsible for preparing reports on development, and for presenting technical seminars both in-house and at conferences.
1983 – 1985: Consultant with IT&A Inverse Theory and Applications Inc., Vancouver, British Columbia, Canada. Responsibilities primarily involved in the development and application of algorithms for specific client contracts, and in the preparation of final reports. Also responsible for the maintenance and application of the Karhunen- Loève seismic data enhancement routines.

PROFESSIONALLY RELATED FIELD EXPERIENCE:

- September - November 1985: Trainee on dynamite, Vibroseis, and marine seismic crews with CGG (Europe).
September 1981: Field assistant in the CO-CRUST Saskatchewan seismic refraction project, Saskatchewan, Canada.
June 1981: Field assistant aboard the CHSS HUDSON during the CO-CRUST NE Pacific refraction experiment: west of the Queen Charlotte Islands, British Columbia, Canada.
August 1978 and 1979: Field assistant conducting induced polarization field surveys over the Cavendish test site, Ontario, Canada.
June - July 1975: Field assistant conducting gravity surveys with the Applied Geophysics Branch of the Institute of Geological Sciences, England.

ACADEMIC HISTORY:

- 2008 – 2023: MSc/PhD supervisor, University of Leeds
2012 – 2023: external (honorary) lecturer, University of Leeds
2008 – 2015: external (honorary) lecturer, Imperial College, London
1987 – 1990; 2012-2015: MSc supervisor, Imperial College, London
1980 - 1985: University of British Columbia; [Ph.D. in Geophysics](#):
Thesis entitled "[Applications of the Karhunen- Loève Transform in Reflection Seismology](#)".
1978 - 1980: University of Western Ontario; M.Sc. in Geophysics: Thesis entitled "Analyses of Global Seismicity".
1977 - 1978: University of Calgary.
1977 - 1985: Teaching assistant, Universities of British Columbia, Western Ontario, and Calgary.
1974 - 1977: Manchester University; B.Sc. joint Honours (2:1) in Physics with Geology.

LANGUAGES: English - native tongue; French - conversational; Spanish – elementary, Dutch - awful.

AWARDS:

- **EAGE best 2025 instructor for the course:**
"State of the Art in Full Waveform Inversion"
- **EAGE best paper award for the 2019 First Break paper:**
"Tutorial: The mechanics of waveform inversion".
- **EAGE 2018 honorary life membership**, for contributions to education in geophysics
- **EAGE 2018-2020 international educational tour instructor**
- **Made honorary 'Chief' of the Igbo Peoples, Nigeria, 2016**, for educational charitable works
- **SEG Honourable Mention, 2014 meeting, for the presentation:**
Seismic imaging in and around salt bodies: problems and pitfalls,
- **ION Ambassador Award 2013.**
For 'promoting the image of ION within the wider geophysical community'
- **SEG Honorary Lecturer 2012.**
For 'contributions to advancing the science and technology of geophysics'
- **EAGE Anstey Medal, 2003.**
'For numerous and excellent contributions to seismic processing and depth imaging'
- **SEG Honourable Mention, for the 2003 TLE publication:**
'Factors affecting frequency content in preSDM imaging'
- **CSEG Best Paper Award, 2002 meeting, for the presentation:**
'Anisotropic Ambiguities in TI Media'
- **SEG Honourable Mention, 1997 meeting, for the presentation:**

‘Continuous 3D preSDM velocity analysis’

• **CSEG Best Student Paper Award, 1985 meeting, for the presentation:**

‘Application of the Karhunen-Loève transform for signal-to-noise ratio enhancement in multichannel seismic data’

PROFESSIONAL AFFILIATIONS & INVOLVEMENT:

- Member of the European Association of Geoscientists and Engineers (EAGE); Society of Exploration Geophysicists (SEG); & Geoscience Energy Society of Great Britain (GESGB)
- Associate Editor, Geophysics, SEG (2009 – 2021)
- Member, SEG editorial board (2008 – 2021)
- Member, SEG HL/DL selection committee (2012 – 2018)
- Member, SEG Continuing Education committee (2009 – present)
- Associate Editor, Geophysical Prospecting, EAGE (2009 – 2021)
- Technical Programme Officer, EAGE technical programme committee (2008 – 2017)
- Member, EAGE Oil & Gas Geoscience Division committee (2008 – 2015)
- Member, University of Leeds MSc Geophysics course external advisory board (2016 – 2023)
- Member, Imperial College London MSc Geophysics course external advisory board (2012 - 2015)

WORKSHOP & COURSE ORGANIZATION

- 2009 – present: instructor of the course ‘An Introduction to Migration and Velocity Model Building’ taught for the EAGE, SEG, CSEG, SPG, and several universities.
- 2024: co-organizer, Cairo EAGE *Data Processing* workshop.
- 2018-2020: EAGE Honorary Lecture tour EET13 ‘Velocities, Imaging, and Waveform Inversion’ (presented at ~35 locations)
- 2016: co-organizer, Dubai EAGE/SPE *Subsalt Imaging* workshop.
- 2013: co-organizer, EAGE workshop *Interpreting the Uncertain Image*,
- 2012: SEG Honorary Lecture tour ‘From imaging to inversion’ (presented at 30 locations)
- 2010: co-organizer, SEG summer research workshop on *Subsalt Imaging*;
- 2008: instructor, SPG 1-day workshop on Depth Imaging
- 2004: instructor, SPG 1-day workshop on Deep Water Imaging
- 2002: co-organizer, EAGE workshop ‘Kirchhoff versus Wave Equation Imaging’
- 2002: co-organizer, EAGE workshop ‘High Resolution Velocity Estimation’
- 1998: co-organizer, SEG-EAEG summer research workshop ‘Depth Imaging of Reservoir Attributes’

BOOKS & BOOK CHAPTERS:

- Jones, I.F., 2025, *A review of contemporary full waveform inversion techniques and their commercial availability*. Confidential report commissioned by industry sponsors. 97 pages.
- Jones, I.F., 2018, *Velocities, Imaging, and Waveform Inversion (The evolution of characterizing the Earth's subsurface)*, EET 13, EAGE, 234 pages.
- Jones, I.F., 2015, *Estimating subsurface parameter fields for seismic migration: velocity model building*. Chapter in: Encyclopedia of Exploration Geophysics. SEG, pp. U1-1-U1-24. Editors: Vladimir Grechka and Kees Wapenaar.
<http://library.seg.org/doi/pdf/10.1190/1.978156803027.entry3>
- Jones, I.F., 2010, *An introduction to velocity model building*, EAGE, ISBN 978-90-73781-84-9, 296 pages.
- Jones, I.F., Bloor, R.I., Biondi, B., Etgen J.T., 2008, *Pre-stack depth migration and velocity model building*, SEG Geophysics Reprints, 25, Series Editor: Michael A. Pelissier, 840 pages
- Pelissier, M.A., Hoeber, H., van de Coevering, N., Jones, I.F., 2007, *Classics of Elastic Wave Theory*, SEG Geophysics Reprints, 24, Series Editor: Daniel A. Ebram, 534 pages.

VIDEO LECTURES:

- EAGE 2022: ‘From Incremental to Transformational Workflows: Contemporary Imaging and What Comes Next’: [1hr lecture](#)
- EAGE EET13 2020: ‘Velocities, Imaging and Waveform Inversion’: six hour online course <https://learninggeoscience.org/local/pages/?id=266>
- EAGE tutorial 2015: ‘Near Surface Model Building’: 20min lecture <https://www.youtube.com/watch?v=vSftWmZpLUU>
- SEG 2012 Honorary lecture: ‘From imaging to inversion’ 1hr lecture <https://seg.org/Education/Lectures/Honorary-Lectures/2012-HL-Jones>

REFEREED PUBLICATIONS: EAGE links: SEG links:

- Jones, I.F., 2026, Tutorial: the mechanics of waveform inversion – beyond simple acoustic. First Break: under review.
- Jones, I.F., 2026. What constitutes good quality data? (Invited contribution in: Expert Answers Q&A Section): The Recorder,
- Mamdouh, A., Pelz, K., Bezdan, S., Angerer, E., Oteleanu, A., Granser, H., Abubakr, A., Sakharov, A., Jones, I.F., 2023. Deep structural imaging in the Vienna Basin. First Break, 41, no.12., 31-35.
- Jones, I.F., 2023. Tutorial: Compressive sensing. First Break, 41, no.7., 31-40.
- Jones, I.F., Calderon-Macias, C. Ong, B., Berranger, I., 2023. Tutorial: least squares migration and full waveform imaging. First Break, 41, no.3., 27 - 35.
- Pearce E., Booth, A., Rost, S., Sava P., Konuk, T., Brisbourne, A., Hubbard, B., Jones, I.F., 2023. A synthetic study of acoustic Full Waveform Inversion to improve seismic modelling in fern. *The Annals of Glaciology*, 1–5, <https://doi.org/10.1017/aog.2023.10>
- Pearce E., Booth, A., Rost, S., Sava P., Konuk, T., Brisbourne, A., Hubbard, B., Jones, I.F., 2023. Characterising Ice Slabs in Firn Using Seismic Full Waveform Inversion. *The Journal of Glaciology, Journal of Glaciology*. 15, <https://doi.org/10.1017/jog.2023.30>
- Jones, I.F., 2021. Tutorial: a geophysical perspective on machine learning. First Break, 39, no.3., 35-42.
- Pavlopoulou, P. and Jones, I.F., 2020, The influence of source wavelet-estimation error in acoustic time domain full waveform inversion, First Break, 38, no. 7, 33-38.
- Jones, I.F., 2019, Tutorial: Time conversion of depth migrated data: part II, TTI preSDM. First Break, 37, no. 12, 35-45

- Jones, I.F, Felicio, R., Vlassopoulou, A., Hagen, C., 2019, Tutorial: tomographic Bayesian uncertainty estimation. *First Break*, **37**, no.9, 37-48
- Jones, I.F, 2019, Tutorial: The mechanics of waveform inversion. *First Break*, **37**, no.5, 31-43.
- Brittan, J., and Jones, I.F, 2019, FWI evolution – from a monolith to a toolkit, *The Leading Edge*, **38**, no.3, 179-184.
- Jones, I.F., Singh, J., Greenwood, S., Chigbo, J., Cox, P., and Hawke, C., 2018, High resolution impedance estimation using refraction and reflection FWI constraints: the Fortuna region, offshore Equatorial Guinea., *First Break*, **36**, no 11, 39-44.
- Jones, I.F., Kobylarski, M., and Brittan, J. 2017, Tutorial: RTM Image Enhancement; bad models and mad physics, *First Break*, **35**, no 3, 31-38.
- Jones, I.F, 2014, Tutorial: migration imaging conditions. *First Break*, **32**, no.12, 45-55.
- Jones, I.F, and Davison, I., 2014, Seismic imaging in and around salt bodies. *SEG Interpretation*, **2**, no.4, SL1-SL20.
- Jones, I.F, 2013, Tutorial: The seismic response to strong vertical velocity change. *First Break*, **31**, no.6., 43-54.
- Jones, I.F, 2013, Tutorial: Transforms, orthogonality, eigenvectors, and eigenvalues. *First Break*, **31**, no.1., 51-61.
- Jones, I.F, 2013, Absorption related velocity dispersion below a possible gas hydrate geobody, *Geophysical Prospecting*, **61**, no. sup.1, 434-445.
- Jones, I.F, 2012, Tutorial: Incorporating near-surface velocity anomalies in pre-stack depth migration models. *First Break*, **30**, no.3, 47-58.
- Leveille, J.P., Jones, I.F., Wang, B., Bevc, D., 2011, Special Section: Subsalt Exploration and Development - Introduction. *Geophysics*, **76**, No. 5, WB1-WB2.
- Leveille, J.P., Jones, I.F., Zhou, Z-Z., Wang, B., Liu, F., 2011, Subsalt Imaging for Exploration, Production and Development: A Review. *Geophysics*, **76**, No. 5, WB3-WB20.
- Jones, I.F, 2010, Tutorial: ray-based tomography. *First Break*, **28**, no.2, 45-52
- Williamson, P., Wang, B., Bevc, D., Jones, I.F., 2010, Full wave-equation methods for complex imaging challenges. *The Leading Edge*, **29**, no.3, 264-268.
- Jones, I.F, 2009, Tutorial: Time conversion of depth migrated data. *First Break*, **27**, no.7, 51-55.
- Jones, I. F., 2008, A modeling study of pre-processing considerations for reverse-time migration: *Geophysics*, **73**, No. 6; T99–T106.
- Fruehn, J.K., I. F. Jones, V. Valler, P. Sangvai, A. Biswal, & M. Mathur, 2008, Resolving Near-Seabed Velocity Anomalies: Deep Water Offshore Eastern India: *Geophysics*, **73**, No.5, VE235-VE241..
- Jones, I. F., 2008, Effects of pre-processing on reverse time migration – a North Sea study: *First Break*, **26**, no.6, 73-80.
- Smith, P., I. F. Jones, D. G. King, P. Sangvai, A. Biswal, & M. Mathur, 2008, Deep Water Pre-Processing: East Coast India: *First Break*, **26**, no.5, 101-107.
- Jones, I.F., Sugrue, M.J., Hardy, P.B., 2007, Hybrid Gridded Tomography. *First Break*, **25**, no.4, 15-21.
- Stewart, P.G., Jones, I.F., Hardy, P.B., 2007, Solutions for deep water imaging, *Geohorizons*, January, 8-22.
- Farmer, P., Jones, I.F., Zhou, H., Bloor, R., Goodwin, M.C., 2006, Application of Reverse Time Migration to Complex Imaging Problems. *First Break*, **24**, no.9, 65-73.
- Jones, I.F., 2005, Anisotropy (in: Expert Answers Q&A Section);, *The Recorder*, **30**, no.4, 8-10.
- Jones, I.F., Christensen, R., Haynes, J. Faragher, J., Novianti, I., Morris, H., Pickering, G., 2005, Multi-Disciplinary Geoscience: The ‘Brenda’ North Sea Development, *The Recorder*, **30**, no.1, 38-43 [re-printed from the 2004 *First Break* article]
- Sugrue, M., Jones, I.F., Evans, E., Fairhead, S., Marsden, G., 2004, Enhanced velocity estimation using gridded tomography in complex chalk, *Geophysical Prospecting*, **52**, no.6, 683-691.
- Jones, I.F., Christensen, R., Haynes, J. Faragher, J., Novianti, I., Morris, H., Pickering, G., 2004: The Brenda Field Development: a Multi-Disciplinary Approach, *First Break*, **22**, no.6, 85-91.
- Lafond, C., Jones, I.F., Bridson, M., Houlléviq, H., Kerdraon, Y., & Peliganga, J., 2003, Imaging Deep Water Salt Bodies in West Africa, *The Leading Edge*, **22**, no.9, 893-896.
- Jones, I.F., Bridson, M.L., & Bernitsas, N.X., 2003, Anisotropic Ambiguities in TI Media. *First Break*, **21**, no.4, 31-37.
- Jones, I.F., 2003, A review of 3D preSDM velocity model building techniques *First Break*, **21**, no.3, 45-58.
- Jones, I.F., Fruehn, J., 2003, Factors affecting frequency content in 3D preSDM imaging, : *The Leading Edge*, **22**, no.2., 128-134.
- Jones, I.F. & Lambaré, G. 2003, Wave equation versus ray based imaging. *First Break*, **21**, no.2, 11-13.
- Robein, E., Jones, I.F., de Bazelaire, E, 2003, High-density high-resolution seismic velocity analysis: where are we and for what benefit? *First Break*, **21**, no.1, 12-14.
- Jones, I.F. & Folstad, P.G., 2002, Continuous high resolution velocity as a 4D attribute. *First Break*, **20**, no.11, 687-694.
- Jones, I.F., Baud, H., 2001, Velocity as an attribute: continuous velocity estimation from 3D preSDM CRP gathers, *First Break*, **19**, no.1, 19-23.
- Jones, I.F., Baud, H., Henry, B., Strachan, A. Kommedal, J., Gainski, M., 2000, The effect of acquisition direction on 3D preSDM imaging. *First Break*, **18**, no.9, 385-391.
- Jones, I.F., Baud, H., Ibbotson, K., Audebert, F., 2000, Continuous 3D preSDM velocity analysis: *The Leading Edge*, **19**, no.3, 263-269.
- Jones, I.F., Ibbotson, K., Henry, B., Strachan, A., Baud, H., 1998, Enhancements to 3D preSDM salt-flank imaging, *Journal of Seismic Exploration*, **7**, no.3/4, 329-346..
- Jones, I.F., Ibbotson, K., Grimshaw, M., & Plasterie, P, 1998: 3D pre-stack depth migration and velocity model building; *The Leading Edge*, **17**, no.7, 897-911. Also published in: 3-D Seismic Exploration: Ed. R.J. Graebner, B.A.Hardage, & W.A.Schneider, Soc. of Expl. Geophys. 2001, 430-438.
- Jones, I.F., Diet, J.P., Guillaume, P., & Audebert, F., 1995, (*invited paper*) A strategic approach to 3D pre-stack depth imaging: proceedings of the 1995 spring symposium of the Geophysical Society of Tulsa, Society of Exploration Geophysicists, 53-69. Also published in: Seismic Depth Estimation, Ed. W.S. Harlan, publ. Geophysical Society of Tulsa.
- Hardy, R.J., Hobbs, R.W, Warner, M.R., and Jones, I.F., 1990, The reliability of multiple suppression: *First Break*, **8**, 297-304.
- Yedlin, M.J., Jones, I.F., and Narod, B.B., 1987, Application of the Karhunen- Loève transform to diffraction separation: *IEEE Transactions on Speech, Acoustics, and Signal Processing*, **ASSP-35**, 2-8.

- Jones, I.F., and Levy, S., 1987, Signal-to-noise ratio enhancement in multichannel seismic data via the Karhunen-Loève transform: **(invited paper)** Geophysical Prospecting, **35**, 12-32.
- Chapman, N.R., Levy, S., Stinson, K., Jones, I.F., Oldenburg, D.W., and Prager, B.T., 1986, Inversion of sound-speed and density profiles in deep ocean sediments: Journal of the Acoustical Society of America, **79**, 1441-1456.
- Jones, I.F., 1985, "[Applications of the Karhunen-Loève Transform in Reflection Seismology](#)". Ph.D. Thesis, 256 pages, The University of British Columbia, Canada.
- Clowes, R.M., Ellis, R.M., Hajnal, Z., and Jones, I.F., 1983, Seismic reflections from the subducting lithosphere?: Nature, **303**, 668-670.
- Ellis, R.M., Spence, G.D., Clowes, R.M., Waldron, D.A., Jones, I.F., Green, A.G., Forsyth, D.A., Mair, J.A., Berry, M.J., Mereu, R.F., Kanasevich, E.R., Cumming, G.L., Hajnal, Z., Hyndman, R.D., McMechan, G.A., and Loncarevic, B.D., 1983, The Vancouver Island seismic project: a CO-CRUST onshore-offshore study of a convergent margin: Canadian Journal of Earth Sciences, **20**, 719-741.
- Jones, I.F., Mansinha, L., and Shen, P.Y., 1982, On the double exponential frequency-magnitude relation of earthquakes: Bulletin of the Seismological Society of America, **72**, 2373- 2375.
- Jones, I.F., 1980: Aspects of global seismicity: M.Sc. thesis, University of Western Ontario, Canada.

OTHER PUBLICATIONS AND PRESENTATIONS (material also presented at various conferences):

- Jones, I., Brittan, J., Chigbo, J., Awobasivwe, G., Osolo, C. and Ukerun, P., 2020. *A historical review of Earth model building*. Africa Oil and Gas Report, 7.
- Calderón Macías, C., Wang, C., Burley, T., Singh, J., Ong, B., and Jones, I.F., 2019. Velocity Uncertainty in LS RTM and Joint Migration and Inversion with the two-way wave equation. SEG W-18: Least-Squares Migration and the Way Forward.
- Jones, I.F., 2018-2020: 'Velocities, Imaging, and Waveform Inversion' EAGE Honorary Lecture tour EET13 (presented at ~37 locations worldwide)
- Jones, I.F., 2012: From Imaging to Inversion, SEG Honorary lecture tour (presented at 30 locations worldwide).
- Jones, I.F., 2010: Migration problems and solutions, key-note address, SPG meeting, Hyderabad.
- Jones, I.F., 2008: Non-uniqueness and ambiguity in velocity model building, key-note address, SPG meeting, Hyderabad.
- Campbell, A.G., Evans, E., Judd, D., Jones, I.F., Elam, S., 2005, A Southern North Sea Multi-Survey preSDM, The 67th annual meeting of the European Association of Exploration Geophysicists.
- Evans, E., Papouin, M., Abedi, S., Gauer, M., Smith, P., Jones, I.F., 2004, Southern North Sea preSDM imaging using gridded tomography, PETEX
- Papouin, M., Gauer, M., Abedi, S., Evans, E., Smith, P., Jones, I.F., 2004, Carboniferous Prospectivity in the Southern North Sea, PETEX
- Jones, I.F. Bridson, M.L., Folstad, P.G., Goodwin, M.C., 2002, 4D modelling and illumination. Proceedings of the 63rd Ann. Internat. Mtg. Europ. Assoc. Expl. Geophys.
- Jones, I.F., Baixas, F., Cambois, G., Manin, M., 2000, The evolution of massively parallel processes in geophysics (*invited paper*). Offshore Magazine. v60, no,7, pp106-109.
- Henry, B., Strachan, A., Jones, I.F., Gainski, M., Kommedal, J., 1998: A multi-survey 3D pre-stack depth migration case history; World Oil, **9**, 53-57.
- Audebert, F., Diet, J. P., Guillaume, P., Jones, I. F. and Zhang, X., 1997, CRP-scans: 3-D preSDM velocity analysis via zero-offset tomographic inversion: Annual Meeting Abstracts, Society of Exploration Geophysicists, 1805-1808.
- Jones, I.F., Grimshaw, M., Ibbotson, K., Jolley, B., 1996, A new technique for 3D preSDM model building - a North Sea case study: The 58th annual meeting of the European Association of Exploration Geophysicists.
- Jones, I.F., 1996, 3D seismic imaging system for oil and gas reservoirs: final report for E.U. funded THERMIE project: contract OG 188/94 FR, Commission of the European Community, Director General for Energie, February.
- Jones, I.F., Diet, J.P., Guillaume, P., & Audebert, F., 1995, A strategic approach to 3D pre-stack depth imaging: The 57th annual meeting of the European Association of Exploration Geophysicists.
- Diet, J.P., Jones, I.F., & Guillaume, P., 1994, Strategic seismic imaging- a stepwise approach to imaging complex structures: The joint EAEG/SEG summer workshop - "Construction of 3D macro velocity-depth models",
- Jones, I.F., Mandache, V., Campbell, S., Lancaster, S., 1994, 3D-AVO processing: evolution of a processing sequence: proceedings of the 56th annual meeting of the European Association of Exploration Geophysicists.
- Jones, I.F., 1993, 3D velocity model building via iterative one-pass depth migration: proceedings of the 63rd annual meeting of the Society of Exploration Geophysicists.
- Jones, I.F., 1992, Comparative anatomy of 3D one-pass depth migration schemes: proceedings of the 62nd annual meeting of the Society of Exploration Geophysicists (omitted from expanded abstracts).
- Doicin, D., and Jones, I.F., 1988, A rationale for attenuation compensation: proceedings of the Norwegian Petroleum Society.
- Levy, S., Ulrych, T.J., Jones, I.F., and Oldenburg, D.W., 1983, Applications of complex common signal analysis in exploration seismology: proceedings of the 53rd annual meeting of the Society of Exploration Geophysicists.
- Ulrych, T.J., Levy, S., Oldenburg, D.W., and Jones, I.F., 1983, Applications of the Karhunen-Loève transformation in exploration geophysics: proceedings of the 53rd annual meeting of the Society of Exploration Geophysicists.